

Technology Stack

Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	

Layer	Technology	Purpose
Frontend	HTML5	Structure of the web application
Frontend	CSS3	Styling and responsive design
Frontend	JavaScript	Client-side interactivity and form validation
Backend	Python	Core programming language
Backend Framework	Flask	Web framework for handling requests and APIs
Database	SQLite	Stores crop data, prediction history, and user information
Machine Learning	Scikit-learn	Crop recommendation and yield prediction models
Data Processing	Pandas	Data cleaning, preprocessing, and analysis
Numerical Computing	NumPy	Mathematical and numerical operations
Data Visualization	Matplotlib	Graph generation and result visualization
Data Visualization	Seaborn	Statistical charts and plots
Model Storage	Joblib / Pickle	Save and load trained ML models
Development Environment	Visual Studio Code	Code development and debugging
Version Control	Git	Source code management
Repository Hosting	GitHub	Project collaboration and version control

Layer	Technology	Purpose
Dataset Format	CSV	Agricultural dataset storage
Operating System	Windows 10/11, Linux	Development and deployment environment
Web Browser	Google Chrome, Microsoft Edge	Access the web application

Stage	User Actions	User Goal	Pain Points	System Features / Opportunities
1. Awareness	Learns about OptiCrop through advertisements, college, or agricultural programs	Find a smart solution to improve crop production	Limited knowledge of AI-based farming solutions	Introduce the benefits of AI-powered crop optimization
2. Registration & Login	Creates an account and logs into the application	Access the platform securely	Complex registration processes	Simple and user-friendly authentication
3. Data Entry	Uploads or enters crop, soil, rainfall, temperature, humidity, and environmental data	Provide accurate farm information	Manual data collection can be time-consuming	Easy data entry forms and dataset support
4. Data Validation	Reviews and submits the entered information	Ensure correct data is analyzed	Incorrect or missing values	Input validation and error checking
5. Data Analysis	Starts the analysis process	Understand crop and environmental conditions	Difficulty interpreting agricultural data	AI analyzes crop-environment relationships
6. Pattern Identification	Views trends and relationships in the data	Identify factors affecting crop production	Hidden patterns are difficult to detect manually	Machine learning identifies production patterns and trends
7. Environmental Insights	Reviews environmental impact and climate conditions	Understand environmental influence on farming	Lack of actionable environmental information	Environmental insights based on analyzed data
8. Crop Recommendation	Reviews suggested crops and farming strategies	Choose the most suitable crop	Uncertainty in crop selection	Personalized crop recommendations
9. Resource Optimization	Plans irrigation, fertilizer, and land usage	Reduce cost and improve efficiency	Resource wastage	Smart resource optimization recommendations